

DT-8 stress-kinase: pocket-water assessment and design

QV.25.086 | Drelvessa Therapeutics SAS | 13 Nov 2025

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Design question and outcome

- Where can substitution be explored without relying on an uncertain pocket-water interpretation?
- Crystallography-based docking identified a solvent-facing substitution opportunity.
- Drelvessa Therapeutics SAS carried that opportunity into its next chemistry cycle.
- The design hypothesis remains subject to experimental testing.

Coordinate and ligand provenance

- The client supplied in-house X-ray co-crystal coordinates for its stress-kinase ligands.
- Docking and hit clustering used those coordinates as the structural reference.
- Input and preparation records are linked to QDATA2025-0522.

Engagement chronology

- Analysis started 08 Sep 2025.
- Interim interpretation: 13 Oct 2025.
- Final design readout and accompanying report: 13 Nov 2025.
- The ranked design rationale belongs to QV.25.086.

Coordinate data acceptance

- Check coordinate identity and the correspondence between modelled ligand and supplied chemistry.
- Review alternate conformations and unresolved regions near the pocket.
- Distinguish observed atoms from preparation choices in the analysis notebook.

Ligand preparation

- Preserve parent identity through protonation and conformer preparation.
- Record stereochemical assumptions rather than silently filling ambiguous assignments.
- Inspect the prepared ligand against the supplied co-crystal geometry.

Reference pose as an anchor

- Use the supplied co-crystal pose to constrain interpretation of docking proposals.
- Do not favour score improvement when it requires an implausible pose departure.
- Keep observed geometry distinct from a predicted design pose.

Docking and hit clustering

- Compare proposed designs under a consistent preparation route.
- Cluster related ideas so the ranked list exposes redundant changes.
- Use docking scores in kcal/mol as model outputs, not measured affinity.

Pocket-water question

- Which design conclusions depend on retaining a water in the prepared pocket?
- An observed water position does not establish its free-energy contribution.
- Avoid treating a water-mediated contact as an assured affinity gain.

Water-retention sensitivity

- Retained-water preparation: examine the geometry of water-mediated contacts.
- Preserve the provenance of observed and added atoms.
- Omitted-water preparation: examine whether the design interpretation changes.
- Flag proposals dependent on a particular preparation choice.

Conserved-water interpretation

- Assess conservation in the supplied coordinate context.
- Coordinate appearance alone does not establish occupancy in every experimental condition.
- Preserve uncertainty where alternative interaction networks are plausible.

Solvent-facing substitution vector

- The identified opportunity points towards solvent-facing substitution.
- Explore that vector in the next chemistry cycle.
- Retain the co-crystal anchor when judging whether a proposed substituent fits the design rationale.

Design priorities

- Prioritise substitutions that test the solvent-facing hypothesis cleanly.
- Keep water-dependent alternatives distinguishable from the central design direction.
- Avoid interpreting the ranked design list as an experimentally established potency order.

Practical chemistry handover

- Drelvessa Therapeutics SAS took the solvent-facing opportunity into its next chemistry cycle.
- Carry pose rationale and water assumptions with the proposed chemistry.
- Confirm actual sample identity before linking future assay results to a modelled design.

What the model cannot settle

- Docking does not resolve the energetic cost of water displacement by itself.
- Substitution may affect properties outside the binding pocket.
- A plausible pose does not establish cellular activity or selectivity.

Experimental decision after synthesis

- Compare the new material with an appropriate parent reference under matched conditions.
- Interpret measured activity alongside sample quality and relevant properties.
- Return discordant observations to both ligand preparation and pocket-water assumptions.

Reproducibility and report handover

- Dr E. Ostlund: docking interpretation and design ranking.
- Nadia Rahmani: coordinate provenance and preparation mapping.
- Dr T. Zawiernicki: final model-gate recommendation.
- Companion report: drelvessa-qv.25.086-docking-report-v1.pdf, 13 Nov 2025.

Final recommendation

- Use the solvent-facing substitution opportunity to guide the next chemistry cycle.
- Keep pocket-water uncertainty explicit in each design rationale.
- Close QV.25.086 with the computational recommendation and await experimental evidence.
- Dr T. Zawiernicki | 13 Nov 2025